The pxTAB package

Predefined table styles and list-driven tables for LaTeX2e

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2026/03/02 – Version 0.1

Abstract

pxTAB provides a small LaTeX2e API to build tables from comma-separated rows or list-style data without writing raw tabular environments. It offers a handful of predefined visual styles (plain, header, striped, boxed, grid) and a set of composed header styles (header-boxed, header-grid, header-striped), together with key–value options for consistent layout control.

Contents

1 Loading the package

```
\usepackage{pxTAB}
```

Package options are forwarded to the internal key family pxTAB and are equivalent to calling \pxTABsetup in the preamble:

```
\usepackage[style=header,headerbgcolor=blue!15]{pxTAB}
```

2 Configuration (keys)

The pxTAB key family controls both visual style and parsing behaviour. Keys can be set at load time or via:

```
\pxTABsetup{<key>=<value>,...}
```

Defaults are chosen to be reasonable for course work and reports. All keys affect subsequent tables until changed again (usual LaTeX scoping rules apply).

Style and colours

- `style` (choice, default `plain`)
  Selects the high-level style: `plain`, `header`, `striped`, `boxed`, `grid`, or one of the composed header styles `header-boxed`, `header-grid`, `header-striped`. The composed styles behave like the corresponding base style (boxed, grid, striped) with an additional coloured header row.

- `headerbgcolor` (token list, default `blue!15`)
  Background colour of the header row in `header` style.

- `headertextcolor` (token list, default `blue!70!black`)
  Text colour used in the header row.

- `altrowbgcolor` (token list, default `blue!5`)
  Background colour for alternating rows in `striped` style.

- `altrowcolor` (meta)
  Alias for `altrowbgcolor`.

- `altrowtextcolor` (token list, default empty)
  Text colour used on alternating rows in `striped` style.

- `headerfont` (token list, default `\bfseries`)
  Font-switch applied to the header row (e.g. `\sffamily\bfseries`).

- `bodyfont` (token list, default empty)
  Font-switch applied to body rows.

- `rulecolor` (token list, default `black!55`)
  Colour used for horizontal and vertical rules (applied via `\arrayrulecolor`).

- `rulewidth` (dimension, default `0.4pt`)
  Thickness of rules (applied via `\arrayrulewidth`).

Style shortcuts

The following meta keys select a style without the `style=` prefix. They take no argument and can be mixed freely with other keys:

```
\pxTABsetup{header-striped, headerbgcolor=teal!20}
```

- `striped` ≡ `style=striped`
- `boxed` ≡ `style=boxed`
- `grid` ≡ `style=grid`

- `header` ≡ `style=header`
- `header-striped` ≡ `style=header-striped`
- `header-boxed` ≡ `style=header-boxed`
- `header-grid` ≡ `style=header-grid`

Quick header setup

These convenience keys combine a style selection with a header background colour in a single key. The colour argument follows standard `xcolor` syntax. Note that `headertextcolor` and `headerfont` retain their current values (they are not reset by these keys).

- `setheader=<colour>`
  ≡ `style=header, headerbgcolor=<colour>`
- `setheader-striped=<colour>`
  ≡ `style=header-striped, headerbgcolor=<colour>`
- `setheader-boxed=<colour>`
  ≡ `style=header-boxed, headerbgcolor=<colour>`
- `setheader-grid=<colour>`
  ≡ `style=header-grid, headerbgcolor=<colour>`

Example:

```
\pxTable[setheader=teal!20]{Name, Grade}{Alice, 9}
```

Geometry and layout

- `tabcolsep` (dimension, default `6pt`)
  Column separation (mirrors the standard `\tabcolsep` parameter).
- `arraystretch` (token list, default `1.2`)
  Row stretch applied via `\arraystretch`.
- `rowstretch` (meta)
  Alias for `arraystretch`.
- `width` (code, default `natural`)
  Width of the table. The default `natural` keeps the table at its natural (fit-content) width; tables are *not* stretched to page width unless you explicitly set `width` or `tabwidth`. The special values `linewidth` and `line-width` are treated as `\linewidth`; any other value is used as-is (e.g. `.8\linewidth` or `120pt`).

- `tabwidth` (meta)
  Alias for `width`.

- `centering` (boolean, default `false`)
  When `true`, inserts a `\centering` before the internal `tabular` or `tabular*`. This only changes the horizontal alignment of the table block (in a paragraph or inside a `table` float); it does not modify the computed width. In particular, `width=\linewidth` still produces a full-width table, but the block is explicitly centred. The key can be set globally with `\pxTABsetup` or per table in the optional `[<options>]` argument of the commands below.

- `fontsize` (token list, default empty)
  Optional font-size command (such as `\small` or `\footnotesize`) applied to the table.

- `size` (meta)
  Alias for `fontsize`.

- `align` (choice, default `c`)
  When `colspec` is not provided, controls automatic column alignment: `l`, `c`, or `r`.

- `alignment` (meta)
  Alias for `align`.

- `colspec` (token list, default empty)
  Explicit `tabular` column specification (e.g. `ll`, `|c|c|`). When omitted, `pxTAB` builds one automatically from `align`. Simple `l`/`c`/`r` specifications are checked against the inferred column count; on mismatch a warning is issued and an automatic specification is used.

Floats and captions

The following keys enable a lightweight float wrapper around any `pxTAB` table command. They can be set at package level or passed in the per-table options:

- `caption` (token list, default empty)
  When non-empty, the table body is wrapped in a `table` environment and `\caption{<caption>}` is emitted above the tabular.

- `label` (token list, default empty)
  Optional label placed immediately after the caption (or, if there is no caption, at the top of the float body). A recommended convention is to use prefixes such as `tab:grades` so that `\ref{tab:grades}` yields the table number in the usual way.

- `float` (token list, default `H`)
  Placement specifier forwarded to the `table` environment (e.g. `htbp`). The default `H` (from the `float` package, loaded automatically by `pxTAB`) pins the table at the exact source position when the float wrapper is activated by `caption`, `label`, or a non-empty `float` value.

Any non-empty combination of `caption`, `label`, or `float` activates the float wrapper; otherwise tables are generated in-place without a surrounding `table` environment. When both `centering=true` and float keys are used, the `\centering` is inserted inside the `table` environment, before the caption and the tabular body.

Parsing

- `rowsep` (token list, default `;`)
  Separator between rows when using list-style data.
- `cellsep` (token list, default `,`)
  Separator between cells inside each row.
- `probe` (boolean, default `false`)
  When `true`, writes diagnostic information to the LaTeX log file during table rendering (style name, colspec, visual settings, etc.). Intended for debugging and development; no effect on output. QA harnesses may override the internal probe hook for automated assertions.
- `unknown` (code)
  Unknown keys are currently ignored silently (reserved for future extension).

Paragraph cells

By default `pxTAB` produces *inline* columns (`l`/`c`/`r`), suitable for short labels and numbers. Paragraph mode enables fixed-width columns that wrap multi-line content, mirroring the equal-width paragraph layout previously available through `pxTBL`.

- `cellmode` (choice, default `inline`)
  Controls the column type used by the automatic colspec builder:
  - `inline` — standard behaviour; columns are simple alignment columns derived from `align` (`l`/`c`/`r`).
  - `paragraph` — each column becomes a paragraph column (`p{...}` or `m{...}`) with equal width, so cell content wraps across multiple lines.

Paragraph mode is *opt-in*; existing documents that do not set this key are unaffected.

- `paragraphcells` (boolean, default `true` when used)
  Convenience alias: `paragraphcells=true` is equivalent to `cellmode=paragraph`, and `paragraphcells=false` to `cellmode=inline`. When used without an explicit value (bare `\pxTABsetup{paragraphcells}`) it defaults to `true`.

- `paragraphcoltype` (choice, default `m`)
  Selects the underlying LaTeX column type used in paragraph mode:
  - `p` — `p{<width>}` columns; cell content is top-aligned within the cell box.
  - `m` — `m{<width>}` columns (requires the `array` package, loaded automatically by `pxTAB`); cell content is vertically centred within the cell box.

  This key is ignored when `cellmode=inline`.

- `paragraph` (meta, no argument)
  Convenience alias for `cellmode=paragraph`. Example: `\pxTABsetup{paragraph}`.

- `verbose` (meta, no argument)
  Legacy alias for `cellmode=paragraph`, retained for backward compatibility. Prefer `paragraph` or the explicit `cellmode=paragraph` form.

Effective width. When paragraph mode is active the per-column width is computed as

$$w_{\text{col}} = \frac{W_{\text{eff}}}{n_{\text{cols}}}$$

where W_{eff} is the *effective table width*:

- If `width` (or `tabwidth`) is set to an explicit dimension (e.g. `0.8\linewidth`, `120pt`), that value is used directly as W_{eff}.

- If `width=natural` (the default), paragraph mode uses `\linewidth` as W_{eff}, since paragraph columns require an explicit dimension. The resulting table occupies the full text width by default.

Horizontal alignment inside each paragraph cell is governed by the `align` key: `l` maps to `\raggedright`, `c` to `\centering`, and `r` to `\raggedleft`.

Interaction with `colspec`. When the user provides an explicit `colspec`, it is used as-is regardless of `cellmode`. In other words, `cellmode=paragraph` only affects the *automatic* colspec builder; advanced users who need heterogeneous column widths or mixed column types should set `colspec` directly.

3 Presets

A *preset* is a named bundle of key–value settings that can be saved once and applied to any number of tables. Presets are stored globally (they survive group boundaries), but their *effect* is local to the current scope, following normal LaTeX scoping rules.

3.1 Preset commands

- `\pxTABsavepreset{<name>}{<keys>}`
  Stores the key–value string `<keys>` under the name `<name>`. If a preset with the same name already exists it is silently overwritten.

- `\pxTABusepreset{<name>}`
  Applies the preset `<name>` in the current scope (equivalent to calling `\pxTABsetup` with the stored keys). If the preset is not defined, a warning is written to the log and the call is ignored.

- `preset` key
  The `preset` key can be used inside any per-table option list. Because `l3keys` processes keys in source order, keys appearing *after* `preset` override preset values:

  ```
  \pxTable[preset=formal, headerbgcolor=red!20]
    {Name, Grade}{Alice, 9}
  ```

  Here `formal` sets `headerbgcolor=blue!15`, but the explicit `headerbgcolor=red!20` overrides it.

3.2 Built-in presets

Three generic presets are always available (registered at package load time):

Preset	Style	Key bundle
`formal`	`header`	`headerbgcolor=blue!15`, `headertextcolor=blue!70!black`, `headerfont=\sffamily\bfseries`, `centering=true`, `arraystretch=1.3`
`zebra`	`header-striped`	`headerbgcolor=blue!15`, `headertextcolor=blue!70!black`, `altrowbgcolor=blue!5`, `headerfont=\bfseries`, `centering=true`
`minimal`	`plain`	`arraystretch=1.0`, `tabcolsep=4pt`

3.3 Usage example

```
% Define a custom preset:
\pxTABsavepreset{my-rubric}{
  style=header-grid,
  headerbgcolor=teal!30,
  cellmode=paragraph,
  width=\linewidth,
  centering=true
}

% Use it:
\pxTable[preset=my-rubric]{Criterion, Comment}
  {Clarity, Clear and well-structured}
  {Depth, Shows critical analysis}

% Override one key:
\pxTable[preset=my-rubric, headerbgcolor=orange!20]
  {Criterion, Comment}
  {Clarity, Clear and well-structured}

% Apply a built-in preset via \pxTABusepreset:
\begingroup
  \pxTABusepreset{formal}
  \pxTable{Name, Grade}{Alice, 9}{Bob, 7}
\endgroup
```

4 Styles by example

This section illustrates the built-in styles using a small example dataset. In all cases we use:

```
\pxTABsetup{tabwidth=.8\linewidth}
```

4.1 Plain

```
\pxTABsetup{style=plain,tabwidth=.8\linewidth}
\pxTable{Name, Age, City}
        {Alice, 20, Barcelona}
        {Bob, 22, Madrid}
```

Name	Age	City
Alice	20	Barcelona
Bob	22	Madrid

4.2 Header

```
\pxTABsetup
  {
    style=header,
    headerbgcolor=blue!15,
    headertextcolor=blue!70!black,
    headerfont=\sffamily\bfseries
  }
\pxTable{Name, Age, City}
        {Alice, 20, Barcelona}
        {Bob, 22, Madrid}
```

Name	Age	City
Alice	20	Barcelona
Bob	22	Madrid

4.3 Striped

```
\pxTABsetup
  {
    style=striped,
    altrowbgcolor=gray!10,
    altrowtextcolor=black
  }
\pxTable{Item, Quantity}
        {Apples, 3}
        {Oranges, 5}
        {Bananas, 2}
```

Item	Quantity
Apples	3
Oranges	5
Bananas	2

In the `striped`, `header-striped`, and `header-grid` styles, row backgrounds are applied with `\rowcolor` so that the colour fills the *entire cell area* across all columns—including any wrapped lines in paragraph mode. This holds for both natural-width and explicit-width tables.

4.4 Boxed

```
\pxTABsetup{style=boxed}
\pxTableFromList
  {Code, Description;
   PX1, First code;
   PX2, Second code}
```

Code	Description
PX1	First code
PX2	Second code

4.5 Grid

```
\pxTABsetup{style=grid}
\pxTableMatrix{2}{3}
  {m11,m12,m13,
   m21,m22,m23}
```

m11	m12	m13
m21	m22	m23

4.6 Composed header styles

The composed styles combine the header row styling with the boxed, grid, or striped variants so that common table layouts can be selected with a single option. The following examples use the same small dataset.

`style=header-boxed`

```
\pxTABsetup
  {
    style=header-boxed,
    headerbgcolor=blue!15,
    headertextcolor=blue!70!black,
    headerfont=\sffamily\bfseries,
    tabwidth=.8\linewidth
  }
\pxTable{Name, Age, City}
        {Alice, 20, Barcelona}
        {Bob, 22, Madrid}
```

Name		Age		City
Alice		20		Barcelona
Bob		22		Madrid

style=header-grid

```
\pxTABsetup
  {
    style=header-grid,
    headerbgcolor=blue!15,
    headertextcolor=blue!70!black,
    headerfont=\sffamily\bfseries,
    tabwidth=.8\linewidth
  }
\pxTable{Name, Age, City}
        {Alice, 20, Barcelona}
        {Bob, 22, Madrid}
```

Name	Age	City
Alice	20	Barcelona
Bob	22	Madrid

style=header-striped

```
\pxTABsetup
  {
    style=header-striped,
    headerbgcolor=blue!15,
    headertextcolor=blue!70!black,
    headerfont=\sffamily\bfseries,
    altrowbgcolor=gray!10,
    altrowtextcolor=black,
    tabwidth=.8\linewidth
  }
\pxTable{Name, Age, City}
        {Alice, 20, Barcelona}
        {Bob, 22, Madrid}
        {Clara, 23, Girona}
```

Name	Age	City
Alice	20	Barcelona
Bob	22	Madrid
Clara	23	Girona

4.7 Paragraph cells

Paragraph mode produces equal-width columns that wrap long text, suitable for rubrics, notes, and narrative content. The following example uses three columns with `cellmode=paragraph` and a fixed table width:

```
\pxTABsetup
  {
    style=plain,
    cellmode=paragraph,
    paragraphcoltype=p,
    width=.8\linewidth
  }
\pxTable
  {Topic, Description, Status}
  {Design, The API was specified using key-value options
   to keep the interface consistent with existing pxTAB
   commands., Complete}
  {Tests, Paragraph regression tests exercise multi-line
   wrapping and interaction with all styles., Passed}
```

Topic	Description	Status
Design	The API was specified using key-value options to keep the interface consistent with existing pxTAB commands.	Complete
Tests	Paragraph regression tests exercise multi-line wrapping and interaction with all styles.	Passed

Paragraph mode works with all styles. The next example combines `header-striped` with vertically-centred `m` columns:

```
\pxTABsetup
  {
    style=header-striped,
    paragraphcells=true,
    paragraphcoltype=m,
    width=.85\linewidth,
    headerbgcolor=blue!12,
    headertextcolor=blue!70!black,
    altrowbgcolor=gray!10,
    altrowtextcolor=black
  }
\pxTable{Criterion, Comment}
  {Readability, This rubric comment wraps inside
   a paragraph cell while striped backgrounds fill
   the entire cell area across all wrapped lines.}
  {Correctness, Another multi-line comment to show
   that composed styles and paragraph columns work
   together seamlessly.}
```

Criterion	Comment
Readability	This rubric comment wraps inside a paragraph cell while striped backgrounds fill the entire cell area across all wrapped lines.
Correctness	Another multi-line comment to show that composed styles and paragraph columns work together seamlessly.

4.8 Centered and floating tables

The new `centering` and float-related keys can be combined with any of the predefined styles. The following example uses `style=header` with global centering:

```
\pxTABsetup
  {
    style=header,
    headerbgcolor=blue!15,
    headertextcolor=blue!70!black,
    headerfont=\sffamily\bfseries,
    centering=true
```

```
  }
\pxTable{Name, Age, City}
        {Alice, 20, Barcelona}
        {Bob, 22, Madrid}
```

Name	Age	City
Alice	20	Barcelona
Bob	22	Madrid

A float table with caption and label can be created by passing the new keys in the options argument:

```
\pxTable[
  style=header,
  caption={Sample grades},
  label=tab:grades,
  float=htbp
]{Name, Grade}
 {Alice, 9}
 {Bob, 7}
```

Table 1: Sample grades

Name	Grade
Alice	9
Bob	7

As usual, the table can later be referenced with `Table~\ref{tab:grades}` in the surrounding text.

5 Command reference

This section summarises the user-facing commands provided by `pxTAB`. All commands respect the key settings described above.

5.1 `\pxTABsetup`

```
\pxTABsetup{<key>=<value>,...}
```

Sets package-wide defaults in the `pxTAB` key family. Usual LaTeX scoping rules apply, so placing a setup call inside a group affects only the tables in that group.

5.2 \pxTable

```
\pxTable[<options>]{<row1>}{<row2>}...{<rowN>}
```

Builds a table from braced row arguments. Each `<row>` is a comma-separated list of cell contents. A typical example is:

```
\pxTABsetup{style=header}
\pxTable{Name, Age, City}
        {Alice, 20, Barcelona}
        {Bob, 22, Madrid}
```

5.3 \pxTableFromList

```
\pxTableFromList[<options>]{<data>}
```

Builds a table from a single list-style string. By default rows are separated by semicolons (`;`) and cells by commas (`,`):

```
\pxTableFromList
  {Name, Age, City;
   Alice, 20, Barcelona;
   Bob, 22, Madrid}
```

The separators can be customised with the `rowsep` and `cellsep` keys.

5.4 \pxTableRow

```
\pxTableRow[<options>]{<row>}
```

Convenience wrapper for single-row tables (headers or small data rows). It behaves as:

```
\pxTableRow[<opts>]{<row>} == \pxTable[<opts>]{<row>}
```

5.5 \pxTableHeadBody

```
\pxTableHeadBody[<options>]{<header>}{<body-list>}
```

The header row is given explicitly as a comma-separated list, while the body is provided as list-style data using the current separators:

```
\pxTableHeadBody
  {Name, Role}
  {Alice, Dev; Bob, QA; Carol, PO}
```

5.6 \pxTableHeadBodyRows

```
\pxTableHeadBodyRows[<options>]
  {<header>}{<row1>}{<row2>}...
```

Variant of the previous command where both header and body rows are given as separate braced arguments:

```
\pxTableHeadBodyRows
  {Name, Score}
  {Anna, 8}
  {Marc, 9}
  {Joan, 10}
```

5.7 \pxTableMatrix

```
\pxTableMatrix[<options>]{<rows>}{<cols>}{<flat-cells>}
```

Builds a table from a flat list of cells, filled row by row. Missing cells are left blank:

```
\pxTableMatrix{2}{3}
  {m11,m12,m13,
   m21,m22,m23}
```

5.8 Quick reference

The following table summarises the most common commands and their arguments; all of them honour the global and per-table keys described in the configuration section.

Command	Arguments	Typical use
`\pxTABsetup`	`{<key>=<value>,...}`	Set global defaults (centering, colours, fl
`\pxTable`	`[<options>]{<row1>}{...}{<rowN>}`	Small to medium tered as explicit row `centering`, `capti` `float`.
`\pxTableFromList`	`[<options>]{<data>}`	Tables given as a string separated by `cellsep`.
`\pxTableRow`	`[<options>]{<row>}`	Single-row tables quick inline tables).
`\pxTableHeadBody`	`[<options>]{<header>}{<body-list>}`	Head/body split wi given as list-style da
`\pxTableHeadBodyRows`	`[<options>]{<header>}{<row1>}{...}`	Head/body split header and body ro explicitly.
`\pxTableMatrix`	`[<options>]{<rows>}{<cols>}{<flat-cells>}`	Matrix-style data wh list is reshaped into

The float-related keys `caption`, `label`, and `float`, together with `centering`, can be provided either globally via `\pxTABsetup` or per table in the `[<options>]` argument.

6 Advanced usage

6.1 Best practices

- Use `style=plain` for compact, in-text tables where visual emphasis is not required; `style=header` works well for small datasets with a clear header row.
- For longer datasets, `style=striped` improves readability, while `boxed` and `grid` are suitable when a strong tabular structure is desired (summaries, rubrics, code lists).
- Prefer the natural-width default together with `centering=true` when the table is narrower than the text block; this keeps columns tight while visually centring the table.
- Use `width=\linewidth` (or a fraction thereof) when the table should align with surrounding paragraphs; in this case `centering` is optional but harmless.

- When using floats, set both `caption` and `label` so that tables integrate with the document's list of tables and can be referenced with `\ref`.

6.2 When to use paragraph mode

Paragraph mode (`cellmode=paragraph`) is designed for tables whose cells contain narrative text, rubric comments, or any content that benefits from line wrapping:

- **Prefer paragraph mode** when one or more columns hold multi-line descriptions, notes, or explanatory text. Typical examples include rubric tables, requirement matrices, and coursework summary tables.
- **Prefer inline mode** (the default) when all columns contain short labels, numbers, or single-line entries. Inline columns produce tighter, more compact tables.
- **Width interaction.** Paragraph mode always requires an effective width to compute per-column dimensions. When `width=natural` (the default), `\linewidth` is used automatically; for narrower tables set an explicit width such as `width=0.8\linewidth`. The `centering` key works identically in paragraph and inline mode—it controls horizontal placement of the table block, not the column widths.
- **Vertical alignment.** Use `paragraphcoltype=p` (default) for top-aligned cells or `paragraphcoltype=m` when rows of unequal height should be vertically centred.

Known limitations.

- Paragraph mode produces *equal-width* columns only; heterogeneous widths require a manual `colspec`.
- An explicit `colspec` overrides the automatic paragraph layout entirely—users who set `colspec` are responsible for choosing appropriate `p{...}` or `m{...}` column types.
- There is no `longtable` integration; page-breaking paragraph tables remain the domain of `pxTBL`'s `\pxTBLlong` command or manual `longtable` usage.

6.3 Explicit column specifications

When `colspec` is set, its value is used directly as the `tabular` column specification:

```
\pxTABsetup{style=plain,colspec=ll}
\pxTable{Col 1, Col 2}
        {Left, also left}
```

Simple `l`/`c`/`r` specifications are checked against the inferred column count; on mismatch a warning is issued and an automatic specification is used.

6.4 Width control

The `width`/`tabwidth` keys allow fixed-width tables:

```
\pxTABsetup{style=plain,width=\linewidth,align=l}
\pxTable{A, B, C}{1, 2, 3}{4, 5, 6}
```

Non-`natural` widths use a `tabular*` layout with automatic extra column separation so that the table fits the requested width.

6.5 Font size and row spacing

```
\pxTable[fontsize=\small,rowstretch=1.05,style=header]
        {Name, Notes}
        {Alice, Compact rows with smaller text}
        {Bob, This checks per-table sizing controls}
```

The `fontsize`/`size` and `arraystretch`/`rowstretch` keys apply only to the current table.

6.6 Troubleshooting

- *Empty or missing tables.* If the first row has no cells, `pxTAB` skips the table and issues a warning. Check that your header row is non-empty and that the separators (`rowsep`, `cellsep`) are correct.
- *Column specification mismatch.* When `colspec` is given but its length does not match the inferred column count, an automatic specification is used instead and a warning is written to the log. Adjust `colspec` or let `pxTAB` derive it from `align`.
- *Float placement warnings.* The `float` key is passed directly to `\begin{table}[...]`. If you use an unsupported placement specifier, LaTeX may report an "Unknown float option" warning; choose a standard value such as `h`, `t`, or `htbp`.
- *Missing caption or label.* A float without `label` still typesets correctly but cannot be referenced with `\ref`. Conversely, a `label` without `caption` is allowed; the table is still wrapped in a `table` environment so that the label has a proper float context.

7 Implementation notes

This subsection describes, at a high level, how pxTAB processes table data and renders it. It is intended for maintainers and advanced users who need to understand the internal pipeline.

Row parsing and trailing empty trimming

Input data (from \pxTable, \pxTableFromList, or the head/body helpers) is stored as a sequence of rows in \l__pxtab_rows_seq. For list-style input, rows are split using the rowsep separator (default ;), and cells within each row using cellsep (default ,).

Before rendering, __pxtab_trim_trailing_empty_rows: recursively removes trailing rows that have no non-blank cells. A row is considered empty if every cell, after splitting by cellsep, is blank. This ensures that trailing semicolons (e.g. A,B;C,D;) do not produce extra blank rows in the output.

Column count and colspec derivation

The column count is taken from the *first* row only: it is the number of cells after splitting by cellsep. If the first row is empty, a warning is emitted and the table is skipped.

The column specification (colspec) is determined as follows:

- If the user provided colspec and it is a simple l/c/r string whose length matches the inferred column count, it is used as-is.
- If the provided colspec length does not match, a colspec-mismatch warning is issued and an automatic colspec is built from the align key.
- If no colspec was given, one is built automatically by repeating the align character (default c) for each column.

For boxed and grid styles, vertical rules are added around the colspec (e.g. |c|c|c|).

Style renderers and shared helpers

Each style has a dedicated renderer (__pxtab_render_table_plain:n, header:n, striped:n, boxed:n, grid:n). All renderers share:

- __pxtab_append_row_to_body:nnnn — appends a row of cells to the body token list, with optional text colour and font.

- `\__pxtab_append_row_styled_to_body:nnnnn` — same, plus optional `\rowcolor` for background.
- `\__pxtab_emit_tabular_with_body:N` — applies visual setup (`tabcolsep`, `arraystretch`, `rulecolor`, etc.) and emits the `tabular` or `tabular*` with the accumulated body.

Style-specific behaviour:

- `plain`: all rows use body font; no row colours or rules.
- `header`: first row gets `headerbgcolor`, `headertextcolor`, `headerfont`; body rows use body font.
- `striped`: odd rows plain; even rows get `altrowbgcolor` and `altrowtextcolor`. Backgrounds are applied via `\rowcolor` so they fill the entire cell area (including wrapped lines in paragraph mode), not just the text extent.
- `boxed`: colspec wrapped with `|`; `\hline` before first row and after last row.
- `grid`: colspec wrapped with `|`; `\hline` before and after every row.
- `header-boxed`: header-style first row (colours and font) with boxed frame rules as in `boxed`.
- `header-grid`: header-style first row together with full grid rules as in `grid`. The header row background uses `\rowcolor`, filling the entire cell area.
- `header-striped`: header-style first row; subsequent body rows alternate plain and striped colours like `striped`, with the striping parity starting at the first body row. As with `striped`, row backgrounds fill the full cell width across all columns and wrapped lines.

8 Integration with P3CTeX

When using the `P3CTeX` document class and its core package `pxCORE`, the `pxTAB` package is loaded as part of the default configuration so that table commands are available without extra setup.

8.1 Branded presets

When `pxTAB` is loaded via the `P3CTeX` class, five branded presets are automatically registered before package option processing, so they can even be used in `\usepackage` options:

```
\usepackage[preset=p3c-header]{pxTAB}
```

The branded presets use the P3CTeX colour palette defined in `pxGDX` (`primaryColor`, `darkColor`, `secondaryColor`, etc.).

Preset	Description
`p3c-header`	Standard branded header table: deep navy header with white text (`style=header`, sans-serif bold, centred).
`p3c-data`	Striped data table with branded colours: navy header plus light cyan alternating rows (`style=header-striped`).
`p3c-rubric`	Assessment rubric: full grid with paragraph cells, dark blue header (`style=header-grid`, `cellmode=paragraph`).
`p3c-minimal`	Clean minimal table: no header styling, tight spacing, neutral rule colour (`style=plain`).
`p3c-exam`	Exam/assessment table: dark blue header with light grey alternating rows, serif bold header, slightly wider row spacing (`style=header-striped`).

Example:

```
\documentclass{P3CTeX}
% pxTAB is loaded automatically; branded presets are available.

\begin{document}
\pxTable[preset=p3c-header]{Name, Grade}{Alice, 9}{Bob, 7}
\end{document}
```

8.2 Standalone loading

When `pxTAB` is loaded outside the `P3CTeX` class (e.g. with `\documentclass{article}`), the branded presets are *not* registered. The package behaves identically to a standalone installation: all generic built-in presets (`formal`, `zebra`, `minimal`) remain available, and users can define their own presets with `\pxTABsavepreset`.

9 Legacy pxTBL commands

The three commands described below provide a positional-argument interface to paragraph-column tables. They predate the key–value API of `pxTAB` and are retained for backward compatibility. For full key–value control, use the `pxTAB` commands instead.

All three commands accept a star (`*`) variant that adds alternating row colours (hard-coded `blue!25!`).

9.1 \pxTBLtop

`\pxTBLtop*[<maxwidth>]{<numcols>}[<gap>]{<headers>}[<prefix>]{<body>}`

Builds a fixed-position table (inside a `table` float) with equal-width paragraph columns. The `<headers>` and `<body>` arguments contain *raw tabular content* (cells separated by `&`, rows by `\\`).

- `*` — starred variant adds `\rowcolors` for alternating rows.
- `[<maxwidth>]` — total table width (default `\linewidth`).
- `{<numcols>}` — number of columns.
- `[<gap>]` — inter-column separator passed as `@{<gap>}` (default `\quad`).
- `{<headers>}` — raw tabular header row.
- `[<prefix>]` — column type: `p` (top-aligned) or `m` (vertically centred); default `p`.
- `{<body>}` — raw tabular body rows.

Example:

```
\pxTBLtop{3}{Header A & Header B & Header C}
  {Row 1 Col A & Row 1 Col B & Row 1 Col C \\
   Row 2 Col A & Row 2 Col B & Row 2 Col C}
```

9.2 \pxTBLlong

`\pxTBLlong*[<maxwidth>]{<numcols>}[<gap>]{<headers>}[<prefix>]{<body>}`

Builds a page-breaking `longtable` with the header row repeated on every page. Arguments are identical to `\pxTBLtop`, but the output uses `longtable` instead of a `table` float.
Example:

```
\pxTBLlong{2}{Name & Score}
  {Alice & 9 \\ Bob & 7 \\ Carol & 8}
```

Known quirk. The `[<gap>]` optional argument is accepted but *silently ignored* by `\pxTBLlong`—the longtable colspec does not include `@{<gap>}` separators.

9.3 \pxTBL

`\pxTBL*[<maxwidth>]{<numcols>}[<gap>]{<headers>}[<prefix>]{<body>}`

Convenience command that takes *comma-separated lists* for `<headers>` and `<body>` (instead of raw tabular content) and delegates to the `longtable` renderer internally. Each body item becomes one row; items are separated by `\hline`.
Example:

```
\pxTBL{3}{Name, Age, City}{Alice, 20, Barcelona, Bob, 22, Madrid}
```

Known quirk. The `[<gap>]` argument is silently ignored (inherited from the underlying `\pxTBLlong` renderer).

9.4 Star variant differences

The star variant semantics differ slightly between commands:

- `\pxTBLtop*` colours *odd* body rows only (`\rowcolors*{1}{blue!25!}{}`).
- `\pxTBLlong*` and `\pxTBL*` colour *all* rows (`\rowcolors*{1}{blue!25!}{blue!25!}`).